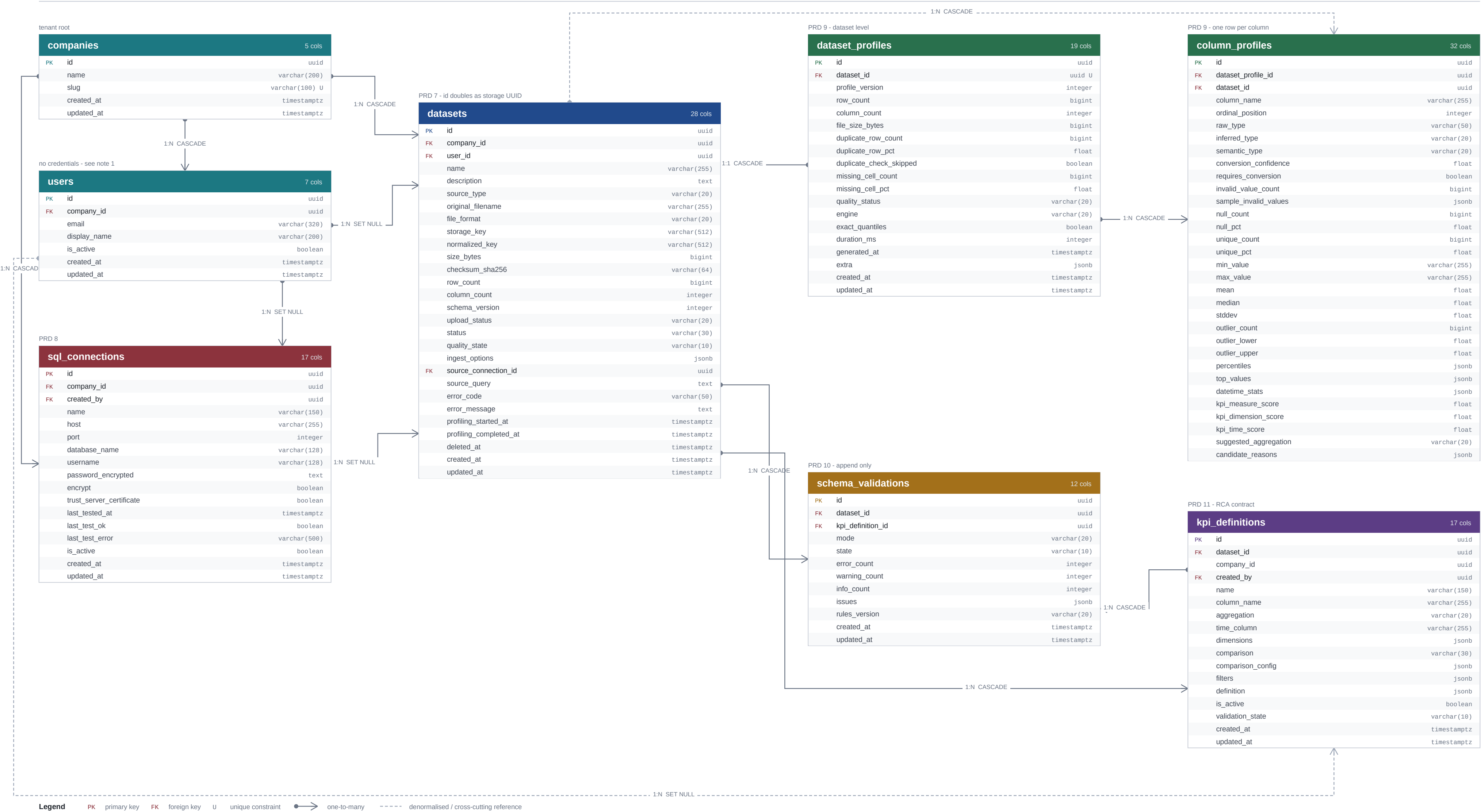


AI Root Cause Analysis Platform

Phase 1 - Data Foundation | entity relationship diagram

8 tables | 13 foreign keys

generated from backend/app/db/models/



Schema notes

Why the model is shaped the way it is

1 users carries no credentials.

Phase 1 ships without authentication by explicit decision, so there is no password, hash or session column anywhere in the schema. The table exists to satisfy the PRD's user_id metadata requirement and to give real auth somewhere to attach later.

2 datasets.id doubles as the physical storage UUID.

The storage key is {company_id}/{YYYY}/{MM}/{dataset_id}.{ext}. The original filename is kept in original_filename as metadata only and never contributes a character to a filesystem path (PRD principle 2).

3 Two different 'status' concepts, deliberately separated.

datasets.status is the pipeline state (pending_upload, uploaded, validating, profiling, profiled, analysis_ready, plus terminal upload_failed / profiling_failed / blocked). dataset_profiles.quality_status is the validation verdict (pass / warning / blocked). The PRD never enumerated either.

4 datasets.quality_state is a denormalised copy.

It mirrors the newest structural schema_validations.state so the dataset list can filter on quality without a join. schema_validations remains the source of truth.

5 column_profiles.dataset_id is denormalised too.

The authoritative parent is dataset_profile_id. Carrying dataset_id as well lets KPI-candidate queries rank columns without joining through dataset_profiles. Both are ON DELETE CASCADE, so the two paths cannot disagree.

6 Per-column statistics are rows, not a JSON blob.

One row per column lets the KPI candidate query rank in SQL and lets the Columns tab paginate server-side. Only the type-specific statistics - percentiles, top_values, datetime_stats, whose shapes differ entirely between numeric, categorical and datetime columns - stay in JSON.

7 kpi_definitions stores the contract twice, on purpose.

The typed columns drive validation and queries; definition holds the frozen, source-agnostic JSON contract handed to the RCA engine in exactly the shape PRD section 11 specifies. The service is the single writer and builds the JSON from the columns.

8 schema_validations is append-only.

Users re-upload corrected files and need to see what changed, so results accumulate rather than being overwritten. mode distinguishes a structural run from one re-checked against a specific KPI definition, which is why kpi_definition_id is nullable.

9 sql_connections has no plaintext password column.

password_encrypted holds a versioned Fernet token ('v1:' prefix) produced by app.core.security. The API read schema has no password field at all, so a credential cannot leak through a response even by mistake.

10 JSON columns are portable across dialects.

Every JSON column is declared JSON().with_variant(JSONB, 'postgres!'). Production keeps JSONB; the test suite runs the identical schema against SQLite.

Evidence + Investigation Layer

Persisted investigations, structured evidence, query trace and audit trail

4 tables | 6 foreign keys

generated from backend/app/db/models/investigation.py

one run - see notes 11, 12

investigations			52 cols
PK	id		uuid
	company_id		uuid
FK	dataset_id		uuid
FK	kpi_definition_id		uuid
FK	created_by		uuid
	question		varchar(500)
	status		varchar(20)
	analysis_state		varchar(30)
	kpi_name		varchar(150)
	measure_column		varchar(255)
	aggregation		varchar(20)
	time_column		varchar(255)
	comparison		varchar(30)
	grain		varchar(20)
	max_drivers		integer
	max_tree_depth		integer
	previous_period_start		timestamptz
	previous_period_end		timestamptz
	current_period_start		timestamptz
	current_period_end		timestamptz
	previous_value		float
	current_value		float
	absolute_change		float
	percentage_change		float
	change_direction		varchar(10)
	severity		varchar(10)
	rows_scanned		bigint
	rows_in_previous_period		bigint
	rows_in_current_period		bigint
	rows_outside_periods		bigint
	queries_executed		integer
	execution_time_ms		integer
	contribution_sum		float
	reconciliation_status		varchar(20)
	reconciliation_tolerance		float
	tree_drift_status		varchar(20)
	evidence_quality		varchar(20)
	evidence_count		integer
	result		jsonb
	tree		jsonb
	decisions		jsonb
	quality_checks		jsonb
	limitations		jsonb
	notices		jsonb
	engine_version		varchar(20)
	source_relation		text
	started_at		timestamptz
	completed_at		timestamptz
	error_code		varchar(50)
	error_message		text
	created_at		timestamptz
	updated_at		timestamptz

one structured claim - see notes 13, 14

investigation_evidence			33 cols
PK	id		uuid
FK	investigation_id		uuid
	company_id		uuid
	sequence		integer
	evidence_type		varchar(30)
	claim		text
	metric		varchar(150)
	dimension		varchar(255)
	dimension_value		varchar(500)
	dimension_value_is_null		boolean
	previous_period		varchar(100)
	current_period		varchar(100)
	previous_value		float
	current_value		float
	absolute_change		float
	percentage_change		float
	contribution_percentage		float
	explanatory_power		float
	filters		jsonb
	source_dataset		varchar(255)
	source_relation		text
	source_columns		jsonb
	query		text
	query_sequence		integer
	analysis_tool		varchar(50)
	validation_status		varchar(20)
	confidence		varchar(10)
	node_id		varchar(500)
	depth		integer
	classification		varchar(20)
	rank		integer
	details		jsonb
	created_at		timestamptz

the real SQL - see note 15

investigation_queries			13 cols
PK	id		uuid
FK	investigation_id		uuid
	sequence		integer U
	purpose		varchar(40)
	sql		text
	parameter_count		integer
	rows_returned		integer
	duration_ms		integer
	status		varchar(10)
	error		varchar(500)
	depth		integer
	node_id		varchar(500)
	created_at		timestamptz

append only - see note 16

investigation_audit_events			9 cols
PK	id		uuid
FK	investigation_id		uuid
	sequence		integer
	event_type		varchar(40)
	message		text
	elapsed_ms		integer
	occurred_at		timestamptz
	details		jsonb
	created_at		timestamptz

11 The execution metrics and verdicts are columns, not JSON.

rows_scanned through evidence_quality are what the Evidence panel shows first, and they are worth filtering on - 'every failed reconciliation this week' should be one indexed query. The findings payload behind them is never queried into, so result, tree, decisions, quality_checks, limitations and notices stay in JSON.

12 reconciliation_tolerance records the value actually applied.

The correctness invariant in app/analysis/rca/constants.py is deliberately not tunable, because a tunable one would let a lost-rows bug be configured into a green tick. The reporting tolerance beside it is configurable, so it is persisted per run: a raised tolerance shows up in the record rather than only in the environment.

13 investigation_evidence.company_id is denormalised.

GET /api/evidence/{id} carries no dataset in its path, so tenant scoping would otherwise need a join through investigations on every lazy-loaded read. The same reasoning as column_profiles.dataset_id in note 5, and both paths are ON DELETE CASCADE.

14 explanatory_power has its own column, never folded into contribution.

Contribution is how much a segment moved the total; explainability is how far the segments deviated from moving in proportion to their size. Movements in opposite directions add to the deviation without adding to the net change, so explainability is not a share and can exceed 100%. Storing them in one column would make that look like a bug.

15 investigation_queries stores the SQL but never the parameters.

The text is the statement handed to DuckDB, verbatim, so 'seven queries in 141 ms' is inspectable rather than merely asserted. Filters and drill predicates bind their values, so a parameter can be a customer name - hence parameter_count and no parameters column, the same reasoning that pins sqlalchemy.engine logging to WARNING.

16 elapsed_ms is the reproducible field, occurred_at is not.

An audit event records a monotonic offset from the run's start, which is identical across two runs of the same data; the wall-clock timestamp is derived from it for display. This mirrors the engine resolving periods against the data's own maximum rather than the clock.

17 Three foreign keys point back to page 1.

investigations.dataset_id is ON DELETE CASCADE - an investigation whose underlying rows are gone can no longer be verified against them, and keeping it would be misleading. kpi_definition_id and created_by are ON DELETE SET NULL, so editing a KPI or removing a user does not destroy the history of what was already found.